

Notes on lecture 14

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1 Change of variables in multiple integrals

In single variable calculus, if we consider $\int_a^b f(x)dx$ and we consider x to be a function of another variable u , say $x = g(u)$, then we have

$$\int_a^b f(x)dx = \int_c^d f(g(u))g'(u)du,$$

where $a = g(c)$ and $b = g(d)$. If we had written $g(u)$ as $x(u)$, then the above equation would have become

$$\int_a^b f(x)dx = \int_c^d f(x(u))\frac{dx}{du}du.$$

Likewise, in double integrals, we are going to make change of variables from (u, v) to (x, y) , defined by the transformation T given by

$$T(u, v) = (x, y),$$

where x and y are related to u and v by the equations

$$x = g(u, v), \quad y = h(u, v).$$

A transformation T is a mapping whose domain and range are both some subsets of $\mathbb{R}^2$.

Suppose the region S in the uv -plane maps to the region R in the xy -plane under the transformation T . If the mapping T is one-one and onto, then the inverse mapping T^{-1} from R to S is well defined. Figure 1.0.1 provides an illustration of the above facts. Under the transformation T^{-1} , say u and v are related to x and y by the equations

$$u = G(x, y), \quad v = H(x, y).$$

Let us consider an example first.

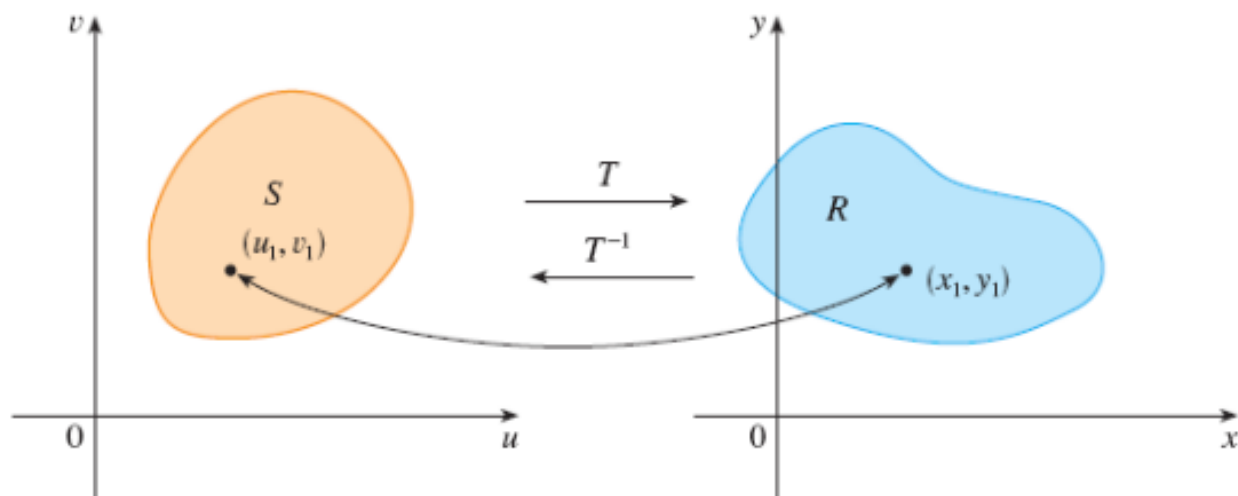


Figure 1.0.1: The transformations T and T^{-1}

Example 1.0.1. Suppose a transformation T is defined by

$$x = u^2 - v^2, \quad y = 2uv.$$

Let $S = \{(u, v) | 0 \leq u \leq 1, 0 \leq v \leq 1\}$. Find the image of the square region S under T in the xy -plane.

The square region S is shown in the left hand side picture of Figure 1.0.2.

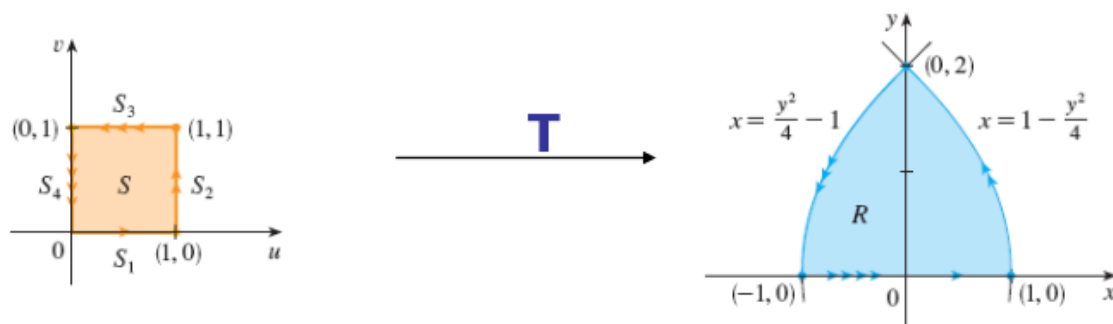


Figure 1.0.2: Figure for Example 1.0.1

Let S_1, S_2, S_3, S_4 denote the boundaries of S (as shown in Figure 1.0.2). Here,

$$S_1 : 0 \leq u \leq 1, v = 0,$$

$$S_2 : u = 1, 0 \leq v \leq 1,$$

$$S_3 : 0 \leq u \leq 1, v = 1,$$

and

$$S_4 : u = 0, 0 \leq v \leq 1.$$

For each $1 \leq i \leq 4$, let $\overline{S}_i$ denote the image of S_i under T in the xy -plane. Remember that the boundaries (S_1, S_2, S_3, S_4) of S in the uv -plane are going to be mapped into boundaries in the xy -plane under the map T .

Observe that on S_1 , we have $x = u^2$ and $y = 0$. Since $0 \leq u \leq 1$ on S_1 , therefore $0 \leq x \leq 1$ on $\overline{S}_1$. So $\overline{S}_1 = \{(x, y) | 0 \leq x \leq 1, y = 0\}$.

On S_2 , we have $x = 1 - v^2$ and $y = 2v$. This implies that $v = \frac{y}{2}$ and hence $x = 1 - \frac{y^2}{4}$. On S_2 , we have $0 \leq v \leq 1$, so $0 \leq y \leq 2$. This gives us a description of $\overline{S}_2$. Similarly, we can find descriptions of $\overline{S}_3$ and $\overline{S}_4$. The right hand side picture of Figure 1.0.2 shows the image R of the square region S under the transformation T . Please see the boundaries of the region R in this picture. $\square$

1.1 How does change of variables affect a double integral?

Let us start with a small rectangle S in the uv -plane, whose lower left corner is the point (u_0, v_0) and whose dimensions are Δu and Δv . Let T be a transformation given by

$$T(u, v) = (x, y),$$

where x and y are related to u and v by the equations

$$x = x(u, v), \quad y = y(u, v).$$

Let the region S transform into the region R in the xy -plane. Let (x_0, y_0) be the image of the point (u_0, v_0) under the map T . Figure 1.1.1 illustrates this.

Let $(u, v) \in S$ transform into the point $P = (x, y) \in R$. Let $\overrightarrow{OP} = \vec{r}$. Then

$$\vec{r} = x\hat{i} + y\hat{j}$$

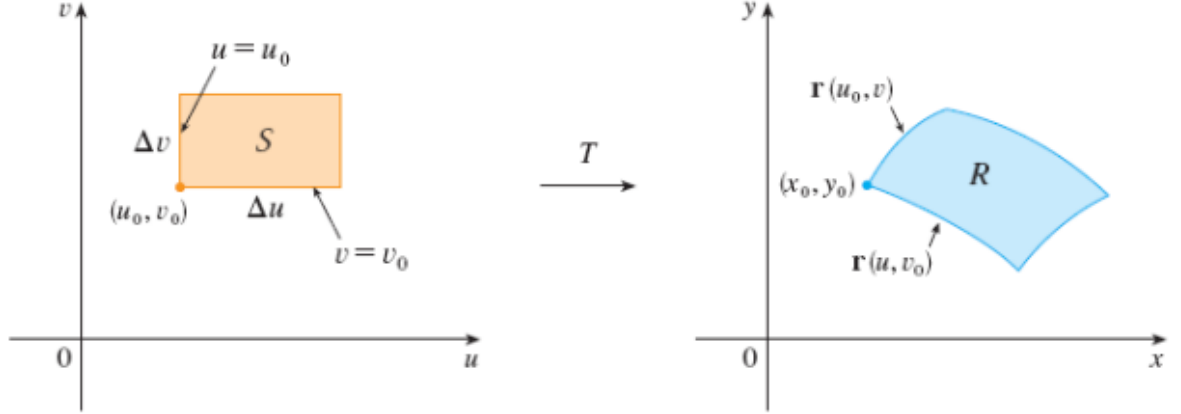


Figure 1.1.1: The image R of the region S under the map T

$$\Rightarrow \vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j}.$$

Let R_1 denote the lower boundary of the curved rectangle R . We know that boundaries map to boundaries under the transformation T . So any vector on the boundary R_1 is going to be of the form

$$\vec{r}(u, v_0) = x(u, v_0)\hat{i} + y(u, v_0)\hat{j}.$$

Since v_0 is fixed for the boundary R_1 , it follows that the curve R_1 has a parametric representation in terms of the single parameter u . So the tangent to this curve R_1 at the point $(x_0, y_0) (= T(u_0, v_0))$ equals $\vec{r}_u(u_0, v_0)$. Similarly, for the left boundary of the curved rectangle R , we have that the tangent to this curve (the left boundary) at the point (x_0, y_0) equals $\vec{r}_v(u_0, v_0)$.

We will now like to approximate the area of the curved rectangle R . Let $\vec{a}$ and $\vec{b}$ be as given in Figure 1.1.2. Then we have

$$\vec{a} = \vec{r}(u_0 + \Delta u, v_0) - \vec{r}(u_0, v_0)$$

and

$$\vec{b} = \vec{r}(u_0, v_0 + \Delta v) - \vec{r}(u_0, v_0).$$

The area of the curved rectangle R can be approximated by the area of the rectangle with sides $\vec{a}$ and $\vec{b}$, which is nothing but $|\vec{a} \times \vec{b}|$. Now,

$$\vec{r}_u(u_0, v_0) = \lim_{\Delta u \rightarrow 0} \frac{\vec{r}(u_0 + \Delta u, v_0) - \vec{r}(u_0, v_0)}{\Delta u}.$$

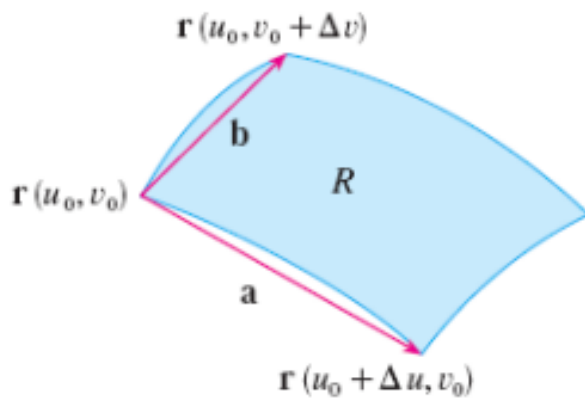


Figure 1.1.2: The vectors $\vec{a}$ and $\vec{b}$

This implies that

$$\vec{r}(u_0 + \Delta u, v_0) - \vec{r}(u_0, v_0) \approx \vec{r}_u \Delta u.$$

That is,

$$\vec{a} \approx \vec{r}_u \Delta u.$$

Similarly,

$$\vec{b} \approx \vec{r}_v \Delta v.$$

Figure 1.1.3 shows the rectangle having sides $\vec{r}_u \Delta u$ and $\vec{r}_v \Delta v$. This implies that

$$|\vec{a} \times \vec{b}| \approx |\vec{r}_u \times \vec{r}_v| \Delta u \Delta v.$$

Recall that

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j}.$$

So we have

$$\vec{r}_u = \frac{\partial x}{\partial u}\hat{i} + \frac{\partial y}{\partial u}\hat{j}$$

and

$$\vec{r}_v = \frac{\partial x}{\partial v}\hat{i} + \frac{\partial y}{\partial v}\hat{j}.$$

$$\Rightarrow \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix}$$

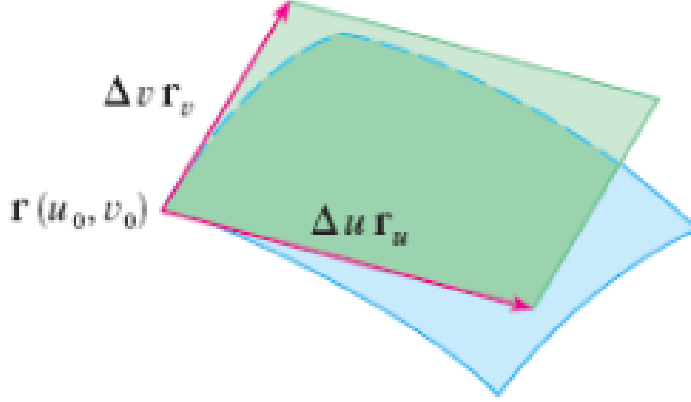


Figure 1.1.3: The rectangle having sides $\vec{r}_u \Delta u$ and $\vec{r}_v \Delta v$

$$= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} \hat{k}.$$

This last determinant $\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix}$ is called the **Jacobian** of the transformation T and it is denoted by $\frac{\partial(x,y)}{\partial(u,v)}$. Observe that

$$\frac{\partial(x,y)}{\partial(u,v)} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}.$$

So

$$|\vec{a} \times \vec{b}| \approx |\vec{r}_u \times \vec{r}_v| \Delta u \Delta v = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| \Delta u \Delta v,$$

where the Jacobian is evaluated at (u_0, v_0) .

Next we divide the region S in the uv -plane into rectangles S_{ij} and call their images in the xy -plane R_{ij} . Figure 1.1.4 illustrates this.

Let us call the area of R_{ij} as ΔA . Then we have

$$\Delta A \approx \left| \frac{\partial(x,y)}{\partial(u,v)} \right| \Delta u \Delta v,$$

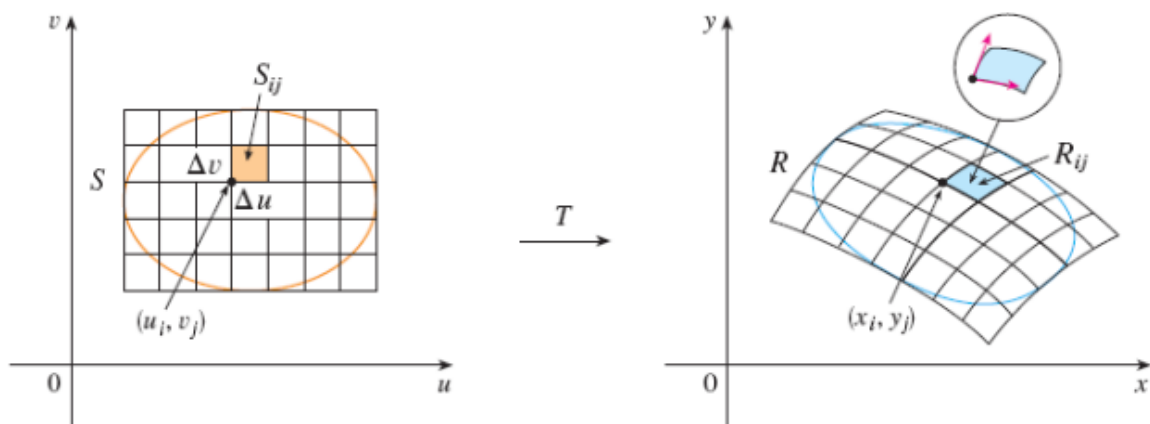


Figure 1.1.4: Dividing S and R into sub-rectangles

where the Jacobian is evaluated at the point (u_i, v_j) . We approximate the double integral of f over R as follows:

$$\begin{aligned} \iint_R f(x, y) dA &\approx \sum_{i=1}^m \sum_{j=1}^n f(x_i, y_j) \Delta A \\ &\approx \sum_{i=1}^m \sum_{j=1}^n f(x(u_i, v_j), y(u_i, v_j)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \Delta u \Delta v \end{aligned}$$

where the Jacobian is evaluated at (u_i, v_j) . Notice that this double sum is a Riemann sum for the integral

$$\iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

The foregoing argument suggests that the following theorem is true. (A full proof is given in books on advanced calculus.)

9 Change of Variables in a Double Integral Suppose that T is a C^1 transformation whose Jacobian is nonzero and that maps a region S in the uv -plane onto a region R in the xy -plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose also that T is one-to-one, except perhaps on the boundary of S . Then

$$\iint_R f(x, y) \, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv$$

Exercise: Please work out Example 3 on slide 15 of Lecture 14.

An Example

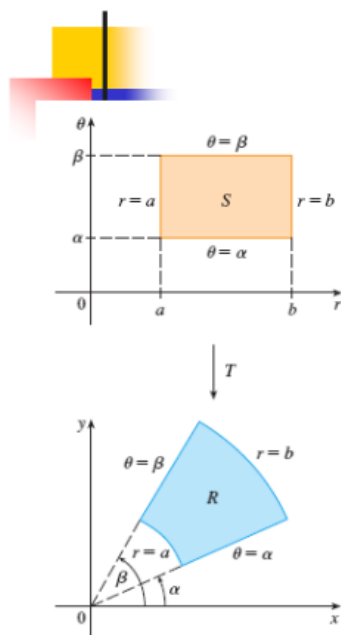


FIGURE 7
The polar coordinate transformation

$$x = g(r, \theta) = r \cos \theta \quad y = h(r, \theta) = r \sin \theta$$

and the geometry of the transformation is shown in Figure 7. T maps an ordinary rectangle in the $r\theta$ -plane to a polar rectangle in the xy -plane. The Jacobian of T is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r > 0$$

Thus Theorem 9 gives

$$\begin{aligned} \iint_R f(x, y) \, dx \, dy &= \iint_S f(r \cos \theta, r \sin \theta) \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| \, dr \, d\theta \\ &= \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta \end{aligned}$$

Example 1.1.1. Evaluate $\int_R \int e^{\frac{x+y}{x-y}} dA$, where R is the trapezoidal region with vertices $(1, 0)$, $(2, 0)$, $(0, -1)$ and $(0, -2)$. The region R has been shown in blue color in Figure 1.1.5 below.

Now R is a region in the xy -plane. We are going to seek the preimage of R in the uv -plane under an appropriate transformation. This is because we are going to perform integration with the u and v variables. We have to define an appropriate transformation. For this, we have to look at the integrand (Looking at the integrand is going to give us a way of getting a transformation.). The transformation need not be unique. We can have more than one transformations, but all of them should give us the same answer finally.

Here, looking at the integrand, we can try the following transformation:

$$u = x + y, \quad v = x - y.$$

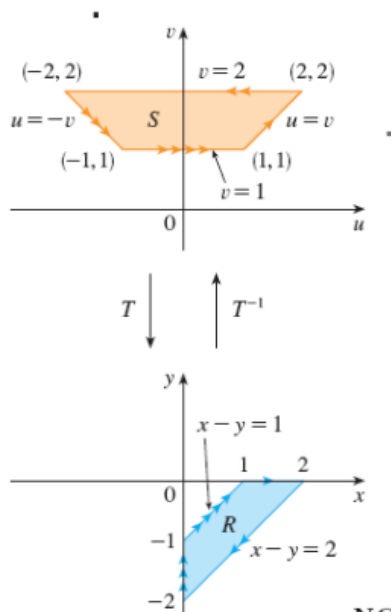


Figure 1.1.5: Figure for Example 1.1.1

Then we have

$$x = \frac{1}{2}(u + v), \quad y = \frac{1}{2}(u - v).$$

Let S denote the preimage of R under the transformation $u = x + y$, $v = x - y$. Let $\overline{S_1}, \overline{S_2}, \overline{S_3}$ and $\overline{S_4}$ denote the 4 boundaries of R , where for each $i \in \{1, 2, 3, 4\}$, $\overline{S_i}$ denoted the boundary of R having i -many arrows marked on it (as shown in Figure 1.1.5). We know that boundaries will map to boundaries under the above transformation. Let S_i (where $i \in \{1, 2, 3, 4\}$) denote the corresponding boundaries of S . Let us first find out the boundaries S_i .

On $\overline{S_1}$, we have $y = 0$ and $1 \leq x \leq 2$. If $y = 0$, then $u = v = x$. When $x = 1$, we have $u = v = 1$. And when $x = 2$, we have $u = v = 2$. So S_1 will be the portion of the line $u = v$ in the uv -plane joining $(1, 1)$ and $(2, 2)$.

On $\overline{S_2}$, we have $y - 0 = \frac{-2-0}{0-2}(x - 2)$. That is, the equation of $\overline{S_2}$ is given by $x - y = 2$. This means that $v = 2$. The line segment $\overline{S_2}$ goes from $(0, -2)$ to $(2, 0)$. So $u (= x + y)$ will vary from $-2 (= 0 + (-2))$ to $2 (= 2 + 0)$. So S_2 is nothing but the portion of the line $v = 2$, where $-2 \leq u \leq 2$.

Similarly, we can find out S_3 and S_4 . Please refer to the portion S in yellow

color shown in Figure 1.1.5. Then

$$\int_R \int e^{\frac{x+y}{x-y}} dA = \int_S \int e^{\frac{u}{v}} \left| \frac{\partial(x, y)}{\partial(u, v)} \right| dudv.$$

Observe that

$$\begin{aligned} \frac{\partial(x, y)}{\partial(u, v)} &= \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \\ &= \frac{1}{2} \left(-\frac{1}{2} \right) - \frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{2}. \end{aligned}$$

Then the above integral becomes $\int_S \int e^{\frac{u}{v}} \frac{1}{2} dudv$.

Observe that S can be considered both as type I and type II regions. But then look at the integrand. If we want to integrate first partially with respect to v , then it is not going to be a straight forward calculation. It is better to integrate partially with respect to u first. So we should consider S as a type II region. Then

$$\begin{aligned} \frac{1}{2} \int_S \int e^{\frac{u}{v}} dudv &= \frac{1}{2} \int_{v=1}^2 \int_{u=-v}^v e^{\frac{u}{v}} dudv \\ &= \frac{1}{2} \int_1^2 \left[v e^{\frac{u}{v}} \right]_{u=-v}^v dv = \frac{1}{2} (e - e^{-1}) \int_1^2 v dv. \end{aligned}$$

The rest of the calculation is left to the reader as an exercise. $\square$

The above example illustrates what order of integration should be followed. In fact, we have to simply look at the integrand and find out which order of integration will make our life easier.

Remark 1.1.2. Sometimes, it may be difficult to transform x and y in terms of u and v . The functions involved in the transformation may be very complicated. In that case, we can use the following property of the Jacobian:

If the inverse transformation exists, then

$$\frac{\partial(x, y)}{\partial(u, v)} \cdot \frac{\partial(u, v)}{\partial(x, y)} = 1.$$

1.2 In case of triple integrals

We are going to consider a region in the uvw -space and this region is going to be transformed into a region in the xyz -space, under some transformation. The formula for the Jacobian as well as for triple integrals will be similar to what we have studied in case of double integrals, except the fact that now we will have to consider a Jacobian in 3 variables, that is, we will have to consider a determinant of order 3. Here, the Jacobian

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

Under similar hypotheses (as used earlier for double integrals), we have the following formula for triple integrals:

$$\int \int \int_R f(x, y, z) dV = \int \int \int_S f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw.$$

1.3 Triple integrals in spherical coordinates

The formula for triple integral in spherical coordinates was:

$$\int \int \int_E f(x, y, z) dV = \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi,$$

where E was a spherical wedge given by

$$E = \{(\rho, \theta, \phi) | a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}.$$

This formula comes from the transformation given by

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

It is your **exercise** to check that the Jacobian $\frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} = \rho^2 \sin \phi$.

Example 1.3.1. Evaluate the integral $\int \int \int_B e^{(x^2+y^2+z^2)^{3/2}} dV$, where $B = \{(x, y, z) | x^2 + y^2 + z^2 \leq 1\}$.

Then under the transformation

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

the region B gets transformed into the region

$$\overline{B} = \{(\rho, \theta, \phi) | 0 \leq \rho \leq 1, 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi\}$$

in the (ρ, θ, ϕ) -space. We also get from the above transformation that $x^2 + y^2 + z^2 = \rho^2$. Then

$$\int \int_B \int e^{(x^2+y^2+z^2)^{3/2}} dV = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} \int_{\rho=0}^1 e^{\rho^3} \rho^2 \sin \phi d\rho d\phi d\theta.$$

The integral on the right hand side of the above equation is in variable separable form. So the above integral becomes

$$= \int_{\theta=0}^{2\pi} d\theta \int_{\phi=0}^{\pi} \sin \phi d\phi \int_{\rho=0}^1 e^{\rho^3} \rho^2 d\rho.$$

It is your exercise to check that this integral equals $\frac{4\pi}{3}(e - 1)$. $\square$

Example 1.3.2. Use spherical coordinates to find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$. Figure 1.3.1 shows the solid in blue color.

Observe that the required volume is nothing but $\int \int_E \int 1 dV$, where E is

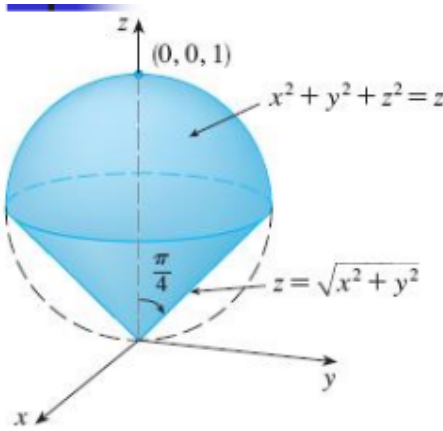


Figure 1.3.1: Figure for Example 1.3.2

the above mentioned solid (as shown in blue color in Figure 1.3.1). Let us use the transformation

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

Under this transformation, the equation $x^2 + y^2 + z^2 = z$ of the sphere becomes

$$\rho^2 = \rho \cos \phi.$$

This implies that

$$\rho = 0, \cos \phi.$$

This means that the lower and upper limit of ρ in the integration are going to be 0 and $\cos \phi$ respectively. So here, we have already obtained the surfaces which are going to bound ρ , namely $\rho = 0$ and $\rho = \cos \phi$. We can see that these are functions of the variable ϕ .

If we consider the cone $z = \sqrt{x^2 + y^2}$, we get (in terms of the transformation) that

$$\begin{aligned} \rho \cos \phi &= \rho \sin \phi \\ \Rightarrow \tan \phi &= 1 \Rightarrow \phi = \frac{\pi}{4}. \end{aligned}$$

So ϕ ranges from 0 to $\frac{\pi}{4}$. One can similarly observe that θ ranges from 0 to 2π . Figure 1.3.2 depicts the ranges of the variables ρ, ϕ and θ .

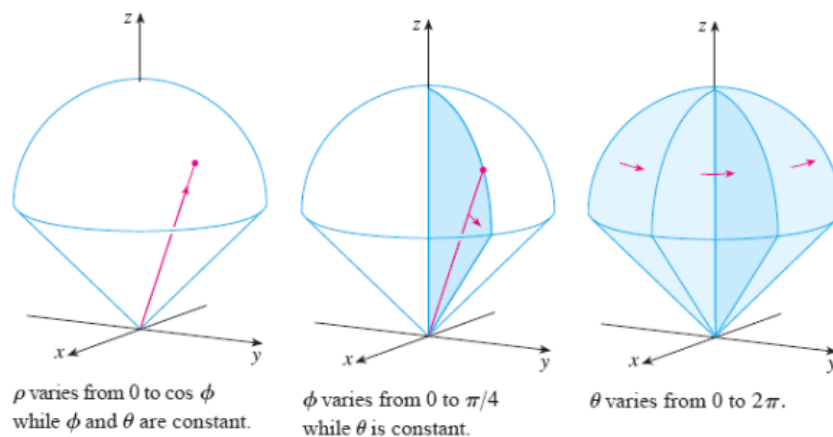


Figure 1.3.2: Another figure for Example 1.3.2

Then we have

$$\int \int_E \int 1 dV = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\frac{\pi}{4}} \int_{\rho=0}^{\cos \phi} 1 \cdot \rho^2 \sin \phi d\rho d\phi d\theta.$$

In the expression on the right hand side of the above equation, we cannot separate out the variables because there is a function (namely, $\cos \phi$) of ϕ in the upper limit of the variable ρ . The above integral becomes

$$= \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\frac{\pi}{4}} \left[\frac{\rho^3}{3} \sin \phi \right]_0^{\cos \phi} d\phi d\theta.$$

The rest of the calculation is left to the reader as an exercise. $\square$

2 Vector fields

We will now start with the new topic “vector fields” and then we will jump to the topic “line integrals” because eventually we are going to do “line integrals of vector functions”. This is going to help us later on in doing surface integrals as well.

Definition 2.0.1. Let E be a subset of $\mathbb{R}^3$. A vector field on $\mathbb{R}^3$ is a function $\overline{F}$ that assigns to each point (x, y, z) in E a three dimensional vector $\overline{F}(x, y, z)$. $\square$

Similarly, one can define a vector field on $\mathbb{R}^2$. Suppose $E \subseteq \mathbb{R}^3$ and for each $(x, y, z) \in E$, we define

$$\overline{F}(x, y, z) = F_1(x, y, z)\hat{i} + F_2(x, y, z)\hat{j} + F_3(x, y, z)\hat{k},$$

then we have defined a vector field $\overline{F}$ on $\mathbb{R}^3$. Here for each $i \in \{1, 2, 3\}$, $F_i(x, y, z)$ is a real valued function of the 3 variables x, y and z . Figure 2.0.1 shows a vector field on $\mathbb{R}^3$. Figure 2.0.2 shows a vector field on $\mathbb{R}^2$.

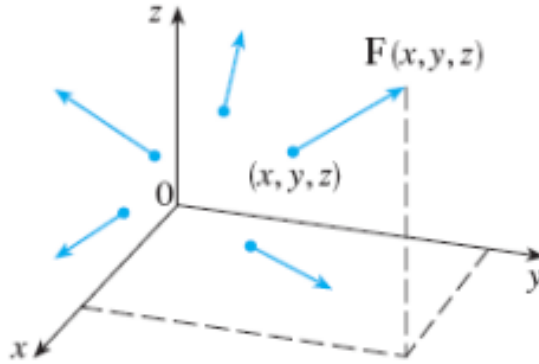


Figure 2.0.1: Vector field on $\mathbb{R}^3$

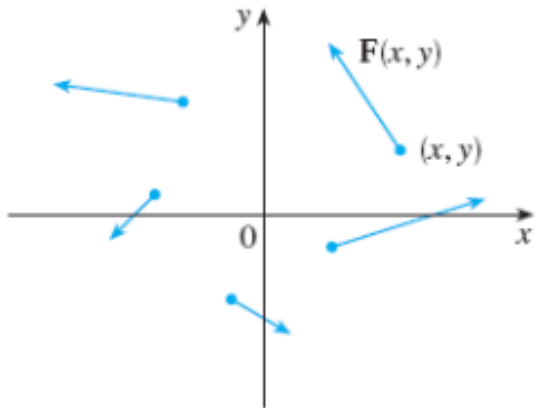


Figure 2.0.2: Vector field on $\mathbb{R}^2$

Example 2.0.2. Let us consider a vector field on $\mathbb{R}^2$ given by

$$\overline{F}(x, y) = -y\hat{i} + x\hat{j}.$$

Suppose we want to define a vector using the above formula at the point $(1, 1)$. Then we have:

$$\overline{F}(1, 1) = -1\hat{i} + 1\hat{j}.$$

So the vector $\overline{F}(1, 1)$ is obtained by moving 1 unit towards the left from the point $(1, 1)$ (reaching the point $(0, 1)$), and then 1 unit vertically up from

the point $(0, 1)$ reaching the point $(0, 2)$. We moved 1 unit towards left from the point $(1, 1)$ because the $-1\hat{i}$ component in the vector $\overline{F}(1, 1)$. Similarly, because of the $1\hat{j}$ component in the vector $\overline{F}(1, 1)$, we had moved 1 unit vertically up from the point $(0, 1)$.

The vector $\overline{F}(1, 1)$ is shown in the xy -plane by joining the point $(1, 1)$ with the point $(0, 2)$ by an arrow pointing towards the point $(0, 2)$. We can draw similar vectors at other points in the xy -plane. Figure 2.0.3 shows some vectors corresponding to this vector field. Figure 2.0.4 shows a complete

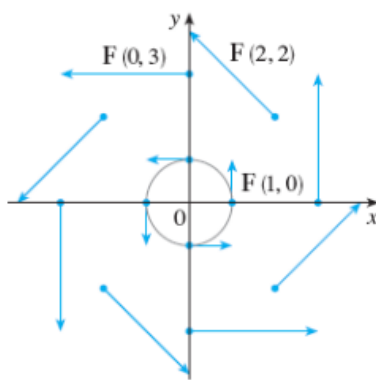


Figure 2.0.3: Figure for Example 2.0.2

picture of the vector field $\overline{F}$. □

Remark 2.0.3. We can draw vector fields on $\mathbb{R}^3$ in a similar way.

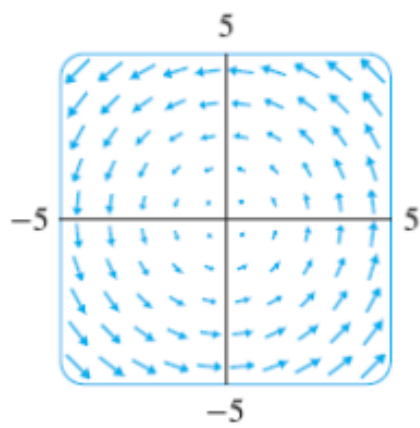


Figure 2.0.4: The vector field for Example 2.0.2

Question: Look at Figure 2.0.5 below. What could be the function defining the vector field corresponding to this figure?

Answer: Observe that in Figure 2.0.5, all the vectors are either pointing

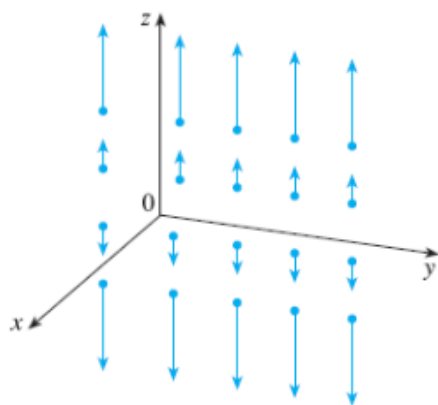


Figure 2.0.5: What is the function defining this vector field?

upwards or downwards, that is, they are in the z -direction. So the x and y components of the corresponding vector field must be 0. The figure also shows that if we increase the value of z , then the length of the vector is also increasing. So it is very likely that the function corresponding to this vector field is given by

$$\vec{F}(x, y, z) = z\hat{k}.$$

Definition 2.0.4. Gradient fields: If f is a scalar function of 2 variables x and y , then we know that its **gradient** ∇f is defined by

$$\nabla f(x, y) = f_x(x, y) \mathbf{i} + f_y(x, y) \mathbf{j}$$

— Therefore ∇f is really a vector field on $\mathbb{R}^2$ and is called a **gradient vector field**. Likewise, if f is a scalar function of three variables, its gradient is a vector field on $\mathbb{R}^3$ given by

$$\nabla f(x, y, z) = f_x(x, y, z) \mathbf{i} + f_y(x, y, z) \mathbf{j} + f_z(x, y, z) \mathbf{k}$$

□

—————-END—————